

T09 B OUTDOOR THERMAL COMFORT | O (MAX : 3 PT)

Intent : Address the thermal comfort needs of project occupants in the exterior spaces of the project.

Summary : This WELL feature requires projects to design outdoor spaces to avoid excessive wind and manage elevated temperatures through shading or other strategies.

Issue : As communities expand, development takes the place of natural features of the landscape. By replacing open spaces and vegetation with structures built of materials that retain heat, a phenomenon known as the urban heat island has developed in the urban landscape.^{1,2} The most prominent feature of an urban heat island is a sustained elevated temperature when compared to the immediately surrounding suburban or undeveloped areas.³ This can lead to adverse health outcomes such as heat-related illness, fatigue, increased breathing and increased heart rate.^{1,4} In addition, high-rise buildings also can concentrate prevailing winds to higher speeds that result in pedestrian discomfort and dangerous conditions.⁵

Solutions : Various strategies can be used to mitigate urban heat island effects and improve outdoor thermal comfort, including adding vegetation, providing shading structures, increasing reflectance of building materials and introducing water features.⁶ The use of trees or vegetation facilitates evapotranspiration and provides shade, contributing to lower surface and air temperatures.¹ Co-benefits of introducing urban vegetation and green spaces include improved air quality, water quality (often via improved stormwater management) and savings in energy.⁷ Shelter canopies address solar radiation and provide relief to thermal stress in extreme temperatures. A shaded surface can be up to 25 °C [45 °F] cooler than the peak temperature of a nearby unshaded surface.⁷ Reducing surface temperatures of sidewalks, roadways and parking lots can be effectively achieved through a mixture of shading and high-albedo paving materials.^{8–10} Finally, water bodies can have a positive effect on microclimates in urban areas by moderating extreme temperatures. Small water bodies such as fountains, pools and ponds can regulate temperature fluctuations and help improve thermal comfort during days of extreme heat.^{11,12} Studies suggest that temperatures around and downwind of water bodies can be 1–2°C [2–4°F] lower compared to surrounding areas.¹² Computational fluid dynamics programs and other modeling tools can be used to predict the thermal conditions and likelihood of excessive wind and the effectiveness of countermeasures.^{5,13}

PART 1 MANAGE OUTDOOR HEAT (MAX : 1 PT)

For All Spaces:

Option 1: Outdoor shading

The following areas (if present) are shaded for more than half of daylight hours each day by tree canopies, awnings, or other structures:

- a. At least 50% of pedestrian pathways and building entrances.
- b. At least 25% of parking spaces (if present).
- c. At least 25% of all plazas, seating areas, exercise facilities with a contiguous area of less than 230 m² and other outdoor areas of congregation.

OR

Option 2: Temperature modeling

For pedestrian pathways and building entrances, parking spaces, and plazas, seating areas, exercise facilities with a contiguous area of less than 230 m² and other outdoor areas of congregation, project provides the following:

- a. Highest expected measure of thermal perception for each month (e.g., highest Physiologically Equivalent Temperature, highest Universal Thermal Climate Index).
- b. If the highest measure of thermal perception is associated with “moderate” (or more severe) heat stress, a list of countermeasures within at least two of the following categories and the expected reduction in heat stress that they provide:
 1. Landscaping and greenery.
 2. Manufactured shading systems (e.g., canopies).
 3. Reflectance of manufactured surfaces (e.g., sidewalks, rooftops).
 4. Water features (e.g., ponds, fountains).

Note :

Interior projects may achieve this part if the grounds of the base building meet the requirements, even if they are outside of the interior project boundary.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 AVOID EXCESSIVE WIND (MAX : 1 PT)

For All Spaces:

A computational fluid dynamic model of the building and any adjacent buildings that takes into account at least one day per season (i.e., per quarter) demonstrates the following:

- a. Winds are not expected to exceed 5 m/s for more than 5% of hours in the year in seating areas or 10% of hours on paths and parking lots.¹⁴
- b. Winds are not expected to exceed 15 m/s on paths, parking lots or seating areas for more than 0.05% of hours in the year.¹⁴

Note :

Interior projects may achieve this part if the grounds of the base building meet the requirements, even if they are outside of the interior project boundary.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 3 SUPPORT OUTDOOR NATURE ACCESS (MAX : 1 PT)

For All Spaces:

Project achieves the following features:

- a. Feature T09β Outdoor Thermal Comfort, Part 1 or Part 2.
- b. Feature M09, Part 2: Provide Nature Access Outdoors.

Note : This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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